Predictive Churn Analysis & Retention Modelling

Telecom subscriber base · 15,000 customers · Ishaan Mankotia

31.9%
churn rate

0.887
ROC-AUC (Gradient Boosting)

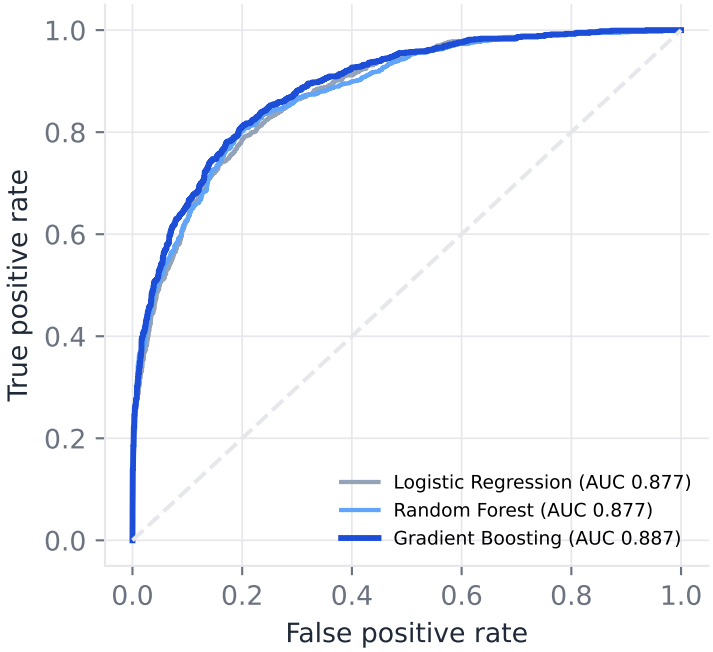
53%
churners in top 20% scored

₹386L
annual revenue at risk

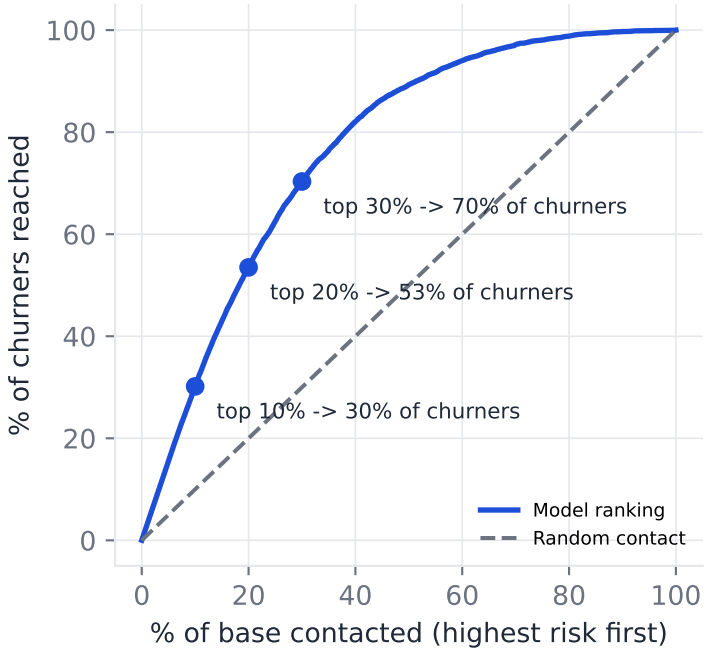
Model comparison (hold-out n = 3,000)

Model	CV AUC	Test AUC	PR-AUC	Precision	Recall	F1
Logistic Regression	0.883	0.877	0.791	0.649	0.779	0.708
Random Forest	0.886	0.877	0.798	0.687	0.742	0.714
Gradient Boosting	0.894	0.887	0.812	0.696	0.750	0.722

ROC curves, hold-out set (n=3,000)



Cumulative gains



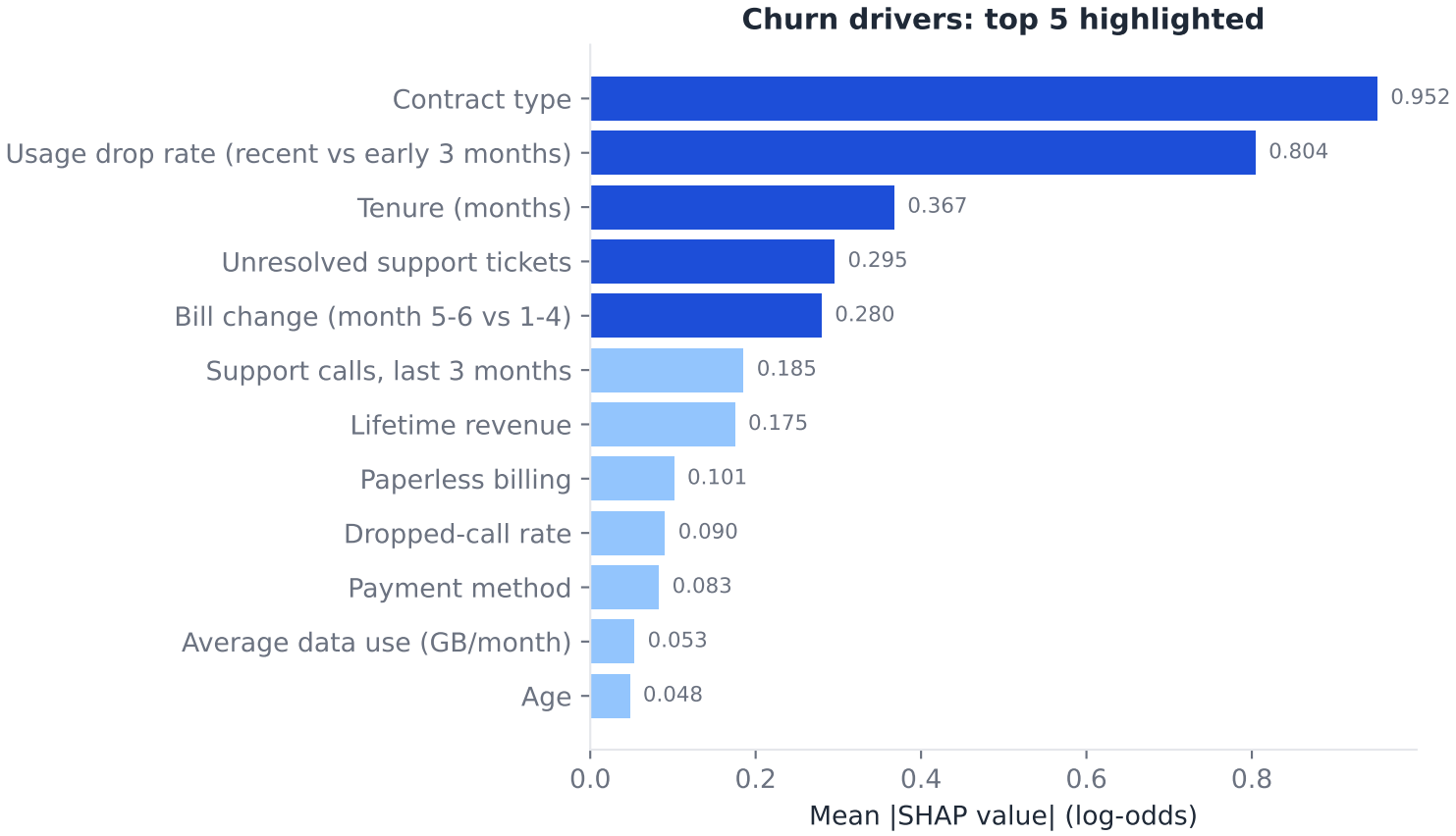
Top 5 churn drivers (SHAP)

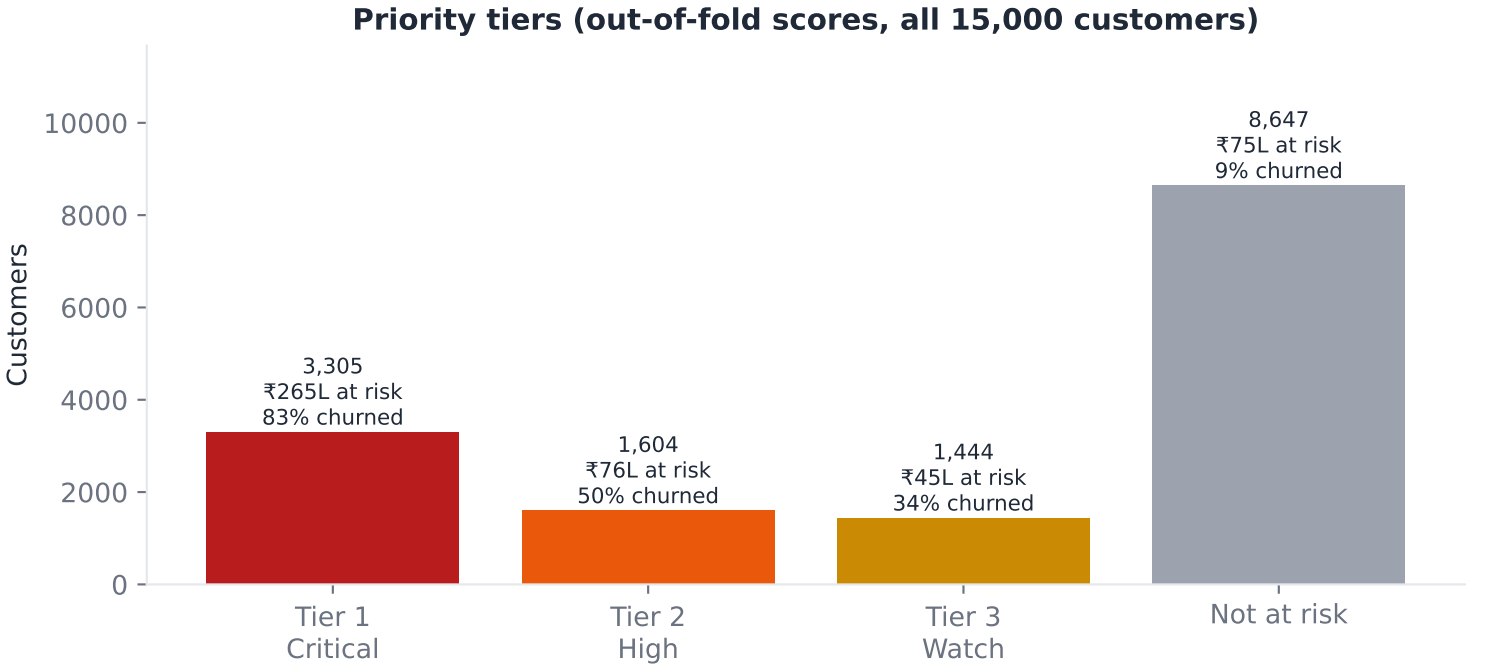
1. Contract type: 25% of attribution, highest risk: Month-to-month, lowest: Two year
2. Usage drop rate (recent vs early 3 months): 21% of attribution, higher -> more churn
3. Tenure (months): 10% of attribution, higher -> less churn
4. Unresolved support tickets: 8% of attribution, higher -> more churn
5. Bill change (month 5-6 vs 1-4): 7% of attribution, higher -> more churn

Recommendations

1. Work Tier 1 first (3,305 customers, 83% actual churn): retention-desk call within 7 days, ordered by revenue at risk (call list in the Excel model).
2. Treat a falling usage trend as the earliest warning: trigger an automated check-in when usage drops 15%+ month on month, before the customer reaches the retention desk.
3. Close the support loop: any customer with a second unresolved ticket goes to a senior agent and gets a callback after resolution. Unresolved tickets are a top-5 driver and are fully within our control.
4. Migrate month-to-month customers to 12-month contracts with a loyalty discount or device bundle; contract type is the largest structural driver.
5. Review the month-5 price rise: 2,068 affected customers churn at 45%. Offer a grandfathered rate to Tier 1-2 customers on Basic and Standard plans.
6. Run every offer with a 10% randomised control group so the save rate is measured, not assumed, and re-score the base monthly.

Drivers and priority tiers





Tier playbook

Tier 1 - Critical: Retention-desk call within 7 days; fix any open ticket first, then offer a 12-month contract with a loyalty discount (up to 15%) or a free plan upgrade.

Tier 2 - High: Targeted offer through the customer's own channel (app/SMS): contract migration incentive for month-to-month users, bill-shock reversal for customers whose bill rose.

Tier 3 - Watch: Automated nurture: usage tips, data top-up bonus, CSAT survey after every ticket; re-score monthly and escalate on any new unresolved ticket.

Synthetic data (seed 42). Methodology: github.com/Ishaan1000/predictive-churn-retention